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Levetiracetam vs. Valproate in a Drosophila Model of Dementia: A Comparative Analysis of Neuroprotection, Neurotoxicity, and Oxidative Stress Pathway Modulation

The Rationale for Investigating Anti-Seizure Medications in Dementia Patients

The investigation of anti-seizure medications (ASMs) within the context of dementia represents a significant shift from their traditional application in epilepsy, driven by compelling clinical observations and emerging pathophysiological understanding. While the initial abstract draft included both epileptic and dementia patients, the user's directive to prioritize dementia patients is scientifically grounded in the high prevalence of subclinical seizures and neuronal hyperexcitability in neurodegenerative diseases, particularly Alzheimer's disease (AD) [111][116]. This focus addresses a critical unmet need, as seizures in AD patients are often unrecognized due to their nonconvulsive nature and may mimic behavioral symptoms of the disease itself [111][187]. Consequently, there is a growing body of evidence suggesting that these undiagnosed epileptiform activities contribute to accelerated cognitive decline and worsen disease progression [116]. Long-term ear-EEG monitoring has revealed that AD patients exhibit a three-fold increased frequency of epileptiform discharges compared to healthy controls, underscoring the widespread nature of this comorbidity [112]. This pathological link provides a strong biological rationale for studying ASMs not merely as anticonvulsants but as potential modulators of the underlying neurodegenerative process.

Beyond seizure control, ASMs are frequently prescribed off-label to manage neuropsychiatric symptoms common in dementia, such as agitation and aggression [109][263]. Gabapentin and pregabalin are among the anticonvulsant drugs noted for having a good safety profile in this context, though caution is advised in cases of severe renal failure [262]. However, this practice is fraught with uncertainty, as robust evidence supporting their efficacy for these indications is lacking, and they carry inherent risks of cognitive

impairment, which can exacerbate the primary pathology [185][238]. This creates a complex clinical dilemma where a treatment intended to improve quality of life may inadvertently accelerate cognitive decline. The situation is further complicated by data suggesting that certain ASMs themselves may increase the risk of developing dementia. A systematic review and meta-analysis identified that the use of valproate, carbamazepine, or clonazepam was associated with a greater risk of incident dementia [36] [41]. Another source concluded that ASM use may be modestly associated with an increased risk of AD in older adults, although the data remain inconclusive [37]. These conflicting reports highlight the urgent need for rigorous preclinical evaluation to dissect the direct effects of individual ASMs on the neurodegenerative cascade, independent of their anti-seizure activity.

The therapeutic potential of some ASMs extends beyond simple symptom management, raising the possibility of direct neuroprotective or even disease-modifying effects. Low-dose levetiracetam, for instance, has been shown in a phase IIa clinical trial to improve spatial memory and executive function in AD patients [110]. It is hypothesized that this benefit stems from its ability to "put the brakes" on hippocampal hyperactivity, a form of neuronal network dysfunction characteristic of early AD [108]. Preclinical studies in mouse models of AD have also demonstrated that levetiracetam, along with lamotrigine and valproic acid, can reduce neuropathology and improve cognitive deficits [214][281]. This suggests that ASMs may act on a shared pathway linking neuronal hyperexcitability to downstream neurodegeneration, making them a compelling class of compounds for investigation. The use of *Drosophila melanogaster* models provides a powerful platform to rapidly screen these drugs and elucidate their mechanisms of action within a genetically tractable system that recapitulates key aspects of human neurodegenerative disease [79] [83].

Comparative Neuroprotective and Neurotoxic Profiles of Anti-Seizure Medications

The decision to administer anti-seizure medications to dementia patients necessitates a careful balancing of potential neuroprotective benefits against significant neurotoxic and cognitive risks. The literature presents a dual perspective on ASMs, highlighting their capacity to either mitigate neuronal damage or exacerbate cognitive decline depending on the specific drug, dosage, and patient population. The neuroprotective potential of certain ASMs is supported by evidence from animal models and early-phase clinical trials.

For example, levetiracetam has demonstrated efficacy in reducing abnormal cortical discharges and reversing memory deficits in a mouse model of Alzheimer's disease [214]. Furthermore, studies in rodent models have shown that levetiracetam, lamotrigine, and valproic acid can all attenuate seizures and improve cognitive outcomes, suggesting a shared mechanism related to normalizing neuronal hyperexcitability [281]. The neuroprotective effect may also be linked to the modulation of pathological protein aggregation; in *Drosophila* models, reduction of endogenous tau expression has been shown to ameliorate neuronal hyperexcitability, a condition targeted by ASMs [72]. By stabilizing neural networks, these drugs may indirectly interrupt the vicious cycle where hyperexcitability promotes further accumulation of toxic proteins like Aβ and tau, leading to a feedback loop of progressive neurodegeneration [8] [177].

Conversely, the neurotoxic and cognitive side effects of ASMs pose a substantial threat to elderly patients with dementia, a population already vulnerable to cognitive impairment. Sedation, somnolence, distractibility, insomnia, and dizziness are among the most commonly reported cognitive side effects of ASM therapy [238]. These effects can significantly impair daily functioning and are a major reason for discontinuation of treatment [185]. The risk is amplified in polypharmacy scenarios; one study found that patients taking a combination of anti-seizure drugs were 2.5 times more likely to experience cognitive dysfunction compared to those on monotherapy [237]. Certain ASMs have raised particular concern. Valproate, while sometimes considered well-tolerated for long-term use in dementia, has been implicated in an increased risk of dementia development in meta-analyses [36] [41] [178]. Similarly, carbamazepine and clonazepam have been associated with higher dementia incidence [36] [41]. Some ASMs, such as gabapentin, are used to try to control agitation and aggressiveness in patients with dementia [109], while others, paradoxically, are associated with increased rates of aggression and irritability in a subset of patients [239][267]. This duality underscores the necessity of a comparative safety assessment to identify agents that offer a favorable risk-benefit profile for this sensitive patient population.

Anti-Seizure Medication	Common Use in Dementia Context	Associated Risks & Side Effects	Cognitive Impact Evidence
Levetiracetam	Seizure control; potential cognitive enhancement at low doses [107][108].	Generally well-tolerated; less sedation than older ASMs [70].	Improved spatial memory and executive function in AD patients [110]; reduces hyperexcitability-induced deficits [214].
Lamotrigine	Treatment of seizures in AD [107][114].	Better tolerated than carbamazepine in the elderly [120].	May be more beneficial for cognition than CBZ or topiramate [113]. Less well-tolerated than levetiracetam after trauma in some dementia patients [118].
Valproate (Valproic Acid)	Off-label for agitation/ aggression (now discouraged) [261].	Increased risk of dementia development [36] [41]. Generally well-tolerated long-term [178].	Associated with increased dementia risk [36].
Carbamazepine	Used in elderly for neuropathic pain, trigeminal neuralgia, and epilepsy [264].	Increased risk of dementia development [36] [41].	
Gabapentin	Management of agitation and aggressiveness [109][262].	Good safety profile, but requires dose adjustment in renal failure [262].	

This comparative analysis reveals a clear stratification of ASMs based on their safety profiles. Levetiracetam and lamotrigine consistently appear in the highest tier for use in older adults with epilepsy, primarily due to their better tolerability and lower incidence of severe cognitive side effects compared to first-generation drugs [70] [120]. Their potential to improve cognition adds another layer of benefit, positioning them as preferred candidates for investigation in dementia. In contrast, valproate, carbamazepine, and clonazepam occupy a lower safety tier, primarily due to their association with an increased risk of dementia and a higher burden of adverse cognitive effects [36] [41] [261]. The stark difference in risk profiles between these drug classes makes them ideal candidates for a comparative study in a preclinical model to validate these differences and uncover the underlying mechanisms, thereby guiding safer and more effective clinical prescribing practices.

Mechanistic Insights into Oxidative Stress Pathways

Oxidative stress stands as a central pillar in the pathogenesis of neurodegenerative diseases, including Alzheimer's disease, and represents a critical mechanistic axis through which anti-seizure medications could exert their diverse effects [20] [21]. It is defined as a pathological imbalance between the production of reactive oxygen species (ROS) and the cell's ability to detoxify them via antioxidant defenses [24] [150]. In the aging brain, which is

already under heightened oxidative pressure, this imbalance becomes exacerbated, contributing directly to neuronal dysfunction and death [147][193]. Evidence from *Drosophila* models confirms that oxidative stress occurs early in AD, preceding the formation of Aβ plaques, and drives neurodegeneration [25][223]. The resulting cellular damage manifests as lipid peroxidation, protein oxidation, and DNA damage, all of which compromise neuronal integrity and function [182]. Key biomarkers of this process, such as malondialdehyde (MDA), a product of lipid peroxidation, and the levels of ROS, are consistently elevated in AD fly models, providing quantifiable readouts for assessing oxidative damage [46][148][191].

Mitochondrial dysfunction is a primary source of ROS in neurodegeneration, creating a self-perpetuating cycle of damage [45][188]. In AD, the accumulation of Aβ peptides has been shown to impair mitochondrial function, leading to reduced synaptic transmission and energy deficits, which in turn fuel further ROS generation [35][173][177]. This mitochondrial-ROS axis is a key driver of the pathology observed in dementia. The cellular response to this oxidative onslaught is orchestrated by a master regulatory pathway: the Keap1/Nrf2 signaling module [55][57]. Under basal conditions, the transcription factor Nrf2 is sequestered in the cytoplasm by its inhibitor, Keap1, which targets it for degradation [102]. Upon exposure to oxidative stress, specific cysteine residues on Keap1 are oxidized, triggering a conformational change that releases Nrf2 [63]. Freed from Keap1, Nrf2 translocates to the nucleus, binds to the Antioxidant Response Element (ARE) in the promoter regions of target genes, and initiates the transcription of a vast array of cytoprotective proteins [207]. These include phase II detoxifying enzymes and antioxidant enzymes such as superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), and heme oxygenase-1 (HO-1) [134][277]. This Nrf2-mediated adaptive response is the cell's primary defense mechanism for maintaining redox homeostasis and protecting against oxidative damage [58].

The significance of this pathway in neurodegeneration is underscored by its genetic links; for instance, mutations in the gene for DJ-1, a recessively inherited cause of Parkinson's disease, disrupt its ability to protect against oxidative stress in a Nrf2-dependent manner [56][61]. Furthermore, DJ-1 is essential for moderating mitophagy, the selective removal of damaged mitochondria, a process crucial for preventing ROS overproduction [100][104]. This intricate connection between mitochondrial health, oxidative stress, and the Nrf2 pathway highlights it as a prime therapeutic target. Several natural products and pharmacological agents have demonstrated neuroprotective effects in *Drosophila* models precisely by activating this pathway, enhancing endogenous antioxidant enzyme activities (SOD, CAT), and reducing markers of oxidative damage like MDA [93][97][168][276]. Therefore,

investigating whether ASMs can modulate the Nrf2-Keap1 pathway offers a direct route to understanding their potential neuroprotective mechanisms. An ASM that boosts the cell's intrinsic antioxidant capacity could counteract the damaging effects of both neurodegenerative pathology and potentially its own side effects, providing a powerful mechanistic explanation for observed phenotypic improvements in locomotor function and survival in a fly model of dementia.

Drosophila melanogaster as a Translational Model for Dementia Research

The fruit fly, *Drosophila melanogaster*, has emerged as an indispensable and highly efficient model system for investigating the complex biology of neurodegenerative diseases, including Alzheimer's disease [79] [218]. Its utility in this research area is rooted in several key advantages that bridge the gap between in vitro studies and costly, time-consuming mammalian models. First, the fly possesses a sophisticated nervous system with functional parallels to that of mammals, allowing for the study of complex behaviors such as learning, memory, and locomotion [232]. These behaviors are readily quantifiable using standardized assays like the negative geotaxis climbing assay, which serves as a sensitive indicator of neurological health and is impaired in various fly models of neurodegeneration [153][159]. Second, the power of *Drosophila* genetics enables researchers to model human disease with remarkable precision. Transgenic techniques allow for the targeted expression of human disease-associated proteins, such as mutant forms of Aβ and tau, directly in the fly brain, successfully recapitulating key pathological hallmarks of AD, including protein aggregation and progressive neurodegeneration [11] [229][233]. These models faithfully mirror many aspects of the human disease, making them predictive platforms for therapeutic discovery [230][270].

Furthermore, the *Drosophila* model is exceptionally well-suited for studying the mechanisms of oxidative stress, a core component of the research goal. The fly shares fundamental metabolic and antioxidant pathways with humans, including the Nrf2/Keap1 ortholog CncC, which regulates a conserved suite of antioxidant genes [55] [210]. This conservation allows for direct investigation of how genetic and pharmacological interventions affect redox homeostasis. Researchers routinely use fly models to assess the neuroprotective effects of compounds by measuring changes in oxidative stress biomarkers like ROS, MDA, and the enzymatic activity of SOD and catalase in fly heads [30] [48] [151]. The ability to perform these biochemical analyses alongside behavioral and

histological assessments in a whole-organism context provides a comprehensive view of a drug's effects. For instance, studies have utilized flies to show that compounds like lutein can increase resistance to oxidative stress and extend healthspan, while neurotoxicants like lead induce oxidative damage that can be measured biochemically [46] [93]. This multi-faceted approach is invaluable for dissecting the complex interplay between neuronal hyperexcitability, oxidative stress, and neurodegeneration.

Finally, the practical advantages of using *Drosophila* make it an ideal platform for high-throughput screening and rapid hypothesis testing. The short lifespan and rapid reproduction rate of flies allow for the assessment of long-term drug effects, such as changes in survival and age-related decline, within weeks rather than months or years [155] [156]. The relatively low cost of maintenance and the vast arsenal of available genetic tools enable researchers to conduct large-scale experiments that would be impractical in vertebrate models [241]. For example, studies have used bang-sensitive *Drosophila* mutants to screen a broad spectrum of ASMs for their ability to suppress seizure activity, demonstrating the model's efficiency in preclinical drug testing [242][243][247]. This capacity for rapid, cost-effective experimentation positions *Drosophila* as a critical translational tool. Findings generated from these fly-based studies can generate strong hypotheses and provide the necessary preliminary data to justify more extensive and expensive investigations in mammalian models and, ultimately, in human clinical trials, thereby accelerating the pace of discovery for novel therapies for dementia.

A Proposed Methodological Framework for In Vivo Assessment

To effectively evaluate the neuroprotective and neurotoxic effects of anti-seizure medications in a *Drosophila melanogaster* model of dementia, a comprehensive methodological framework must be established. This framework should integrate behavioral, physiological, biochemical, and molecular assays to provide a holistic assessment of drug impact. The foundation of the study would be a well-characterized transgenic fly model of Alzheimer's disease. Models expressing human Aβ42 or mutant forms of tau are particularly relevant, as they develop progressive neurodegeneration, locomotor deficits, and reduced lifespan, phenotypes that closely mimic aspects of the human disease [11] [84] [283]. These models serve as a robust platform to test the efficacy of candidate drugs.

The primary behavioral endpoint would be the assessment of locomotor function, typically measured using a negative geotaxis climbing assay [153]. In this test, flies are tapped down to the bottom of a vial, and their ability to climb upwards is recorded over time. Age-related decline in climbing ability is a hallmark of neurodegeneration in AD fly models, and compounds with neuroprotective properties are expected to preserve this function [79] [249]. Assessments would be performed regularly throughout the lifespan of the flies to generate longitudinal data on neurological health. Complementing the behavioral analysis, a lifespan assay is essential to gauge the overall toxicity and survival-promoting potential of the treatments [153] [155]. Survival curves will be plotted and analyzed statistically to determine if any ASM significantly alters the life expectancy of the diseased flies.

To gain deeper mechanistic insights, a series of biochemical assays must be conducted on fly heads, which contain the brain tissue. The level of oxidative stress will be quantified by measuring two key parameters: the production of reactive oxygen species (ROS) and the extent of lipid peroxidation. ROS levels can be detected using fluorescent probes like dihydroethidium (DHE) or MitoSOX, which become fluorescent upon oxidation by superoxide, allowing for visualization and quantification [199] [200] [201]. Lipid peroxidation, a consequence of oxidative damage to cell membranes, will be assessed by measuring the concentration of malondialdehyde (MDA), a stable end-product of this process [46] [148] [151]. Concurrently, the activity of key antioxidant enzymes—superoxide dismutase (SOD) and catalase (CAT)—will be measured [29] [30] [48]. A neuroprotective drug might be expected to decrease ROS and MDA levels while increasing the activity of SOD and CAT, indicating a bolstered antioxidant defense. Finally, to investigate the activation of the Nrf2/CncC pathway, molecular techniques such as Western blotting or quantitative PCR can be employed to analyze the expression levels and nuclear translocation of CncC, the fly homolog of Nrf2, in treated versus untreated groups [202] [204]. Together, these assays create a powerful, multi-tiered approach to dissect the complex effects of ASMs, linking observable phenotypes like behavior and survival to underlying molecular and biochemical mechanisms.

Synthesis and Clinical Implications for Drug Selection

This comprehensive analysis establishes a clear and compelling rationale for utilizing a *Drosophila melanogaster* model to evaluate the neuroprotective and neurotoxic effects of anti-seizure medications in the context of dementia. The convergence of several factors—

the high prevalence of subclinical seizures in AD patients, the common off-label use of ASMs for behavioral symptoms, and the emerging evidence that some ASMs may directly influence cognitive function—creates a critical need for preclinical research to guide safer and more effective clinical practice [109][110][116]. The *Drosophila* model, with its validated AD and tauopathy models and its suitability for assessing oxidative stress, provides an unparalleled platform for this task [55] [79] [218]. By focusing on the central role of oxidative stress and the Nrf2-Keap1 pathway, the proposed research can move beyond simple phenotypic observation to uncover the fundamental mechanisms driving the divergent effects of different ASMs.

Based on the extensive review of the literature, a clear strategy for drug selection emerges. The study should employ a comparative design pitting a low-risk, potentially neuroprotective ASM against a high-risk agent with known associations with cognitive decline and dementia incidence. **Levetiracetam** stands out as the ideal candidate for the low-risk group. It is consistently cited as having the best long-term tolerability profile in elderly epilepsy patients, is well-tolerated in AD populations, and has demonstrated promising cognitive benefits in clinical trials [70] [110][213]. Its mechanism of calming hippocampal hyperactivity aligns perfectly with the therapeutic hypothesis for treating AD-related network dysfunction [108]. Conversely, **valproate** represents the high-risk candidate. Despite some reports of its general tolerability, multiple meta-analyses have linked its use to an increased risk of developing dementia, and it is no longer recommended for managing agitation in dementia patients [36] [41] [261]. Comparing these two drugs will allow for a stark and informative contrast, maximizing the potential to observe clinically relevant differences in neurotoxicity and oxidative stress modulation.

The successful execution of this research program holds significant clinical implications. The findings will provide much-needed preclinical evidence to support or challenge current off-label prescribing practices for ASMs in dementia. If the study demonstrates that levetiracetam mitigates oxidative stress, preserves climbing ability, and extends lifespan in an AD fly model, it would provide a strong biological rationale for its continued use and further exploration for cognitive benefits. Conversely, if valproate is found to exacerbate oxidative damage and accelerate neurodegeneration in the same model, it would reinforce clinical guidelines advising against its use in this population. Ultimately, this research aims to translate complex molecular interactions into actionable clinical knowledge. By identifying which drugs are neuroprotective and which are neurotoxic, and by elucidating the underlying mechanisms involving oxidative stress pathways, this work can help clinicians make more informed decisions, optimize therapeutic strategies, and improve the quality of life for patients with dementia.

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